

## TECHNICAL AND COMMERCIAL PROPOSAL

Assessment, Investigation and Detailed Design of the  
**Existing HVAC Systems — Bakery & Oil Building**

**STAGE 1 : SITE SURVEY & ASSESSMENT | STAGE 2 : ENGINEERING & DETAILED DESIGN**

|                       |                                                                |
|-----------------------|----------------------------------------------------------------|
| <b>Submitted to</b>   | Zad Holding Company Q.P.S.C.                                   |
| <b>Attention</b>      | Mr. Vishnu Prasad (vishnu@qfm.qa)                              |
| <b>Project</b>        | Assessment, Rectification and Upgrade of Existing HVAC Systems |
| <b>Facilities</b>     | Existing Bakery Factory and Oil building, Doha — Qatar         |
| <b>Reference</b>      | MES/2026/HVAC-0137                                             |
| <b>Date</b>           | 06 September 2026                                              |
| <b>Offer validity</b> | 60 days from the date of this proposal                         |
| <b>Basis of offer</b> | Detailed Scope of Work REV-1 and Tender Circular-1             |

## 1. Introduction and Understanding of the Requirement

Metri Engineering Services W.L.L. (MES) is pleased to submit this technical and commercial proposal for the assessment and detailed engineering of the existing HVAC systems serving your bakery factory and QFI Store facilities.

We understand that your Operations team has reported high temperatures in air-conditioned areas and excessive heat in production areas served mainly by ventilation. We also understand and accept that no existing HVAC drawings, equipment schedules, load calculations, air-balance reports or control documentation are available. This proposal is therefore prepared on a complete brownfield basis: MES will establish the full existing condition from site by physical survey, measurement and testing.

In line with Tender Circular-1 (Query 6), this offer covers Stage 1 and Stage 2 under a single lump-sum LPO. Stage 3 implementation is not priced here and will be proposed separately once you have approved the assessment findings and the detailed design.

### Our approach in plain terms

| Step | What it means for you                                                                          |
|------|------------------------------------------------------------------------------------------------|
| 1    | We find out exactly what equipment you have, where it is, and what area each unit serves.      |
| 2    | We measure how each system is actually performing today, under real production conditions.     |
| 3    | We calculate what each room genuinely needs — AC, fresh air and air balance.                   |
| 4    | We identify the real cause of every complaint area, supported by measured evidence.            |
| 5    | We tell you what to repair, what to modify and what to add — with drawings, a BOQ and a price. |

## 2. Stage 1 — Site Survey, Testing and Baseline Documentation

Stage 1 establishes the facts. At the end of Stage 1 you will own a complete, accurate and permanent record of your existing HVAC installation — something that does not exist today.

### 2.1 What we will do

- Prepare and submit an investigation plan showing survey areas, measurement points, instruments, access requirements, safety controls and personnel, for your approval before we mobilise.
- Physically inspect and tag every HVAC, ventilation and exhaust unit within the surveyed areas, and photograph every nameplate. We will record make, model, serial number, capacity, refrigerant, electrical data, year and condition.
- Record which area each unit actually serves, so that every room is linked to the equipment that conditions it.
- Measure room dimensions, heights, doors, shutters, openings and leakage paths, and locate all ovens, proofers, mixers, conveyors, packaging lines and electrical panels.
- Inspect condition: filters, coils, drain pans, fans, belts, bearings, motors, dampers, valves, strainers, insulation, ductwork, controls and sensors.
- Install data loggers and record dry-bulb temperature, relative humidity and dew point by zone over three representative production days at intervals not exceeding 10 minutes, together with matching outdoor conditions.
- Test the actual performance of the equipment: supply and return airflow, external static pressure, entering and leaving air conditions, chilled water flow and temperature difference, refrigerant condition where a deficiency is suspected, and electrical current.
- Measure the complete air balance — supply, return, fresh, transfer and exhaust air — and the pressure relationship between production zones and outside.
- Measure work-zone conditions in hot areas using heat-stress criteria, not dry-bulb temperature alone.
- Interview Operations, Maintenance, QA/QC and HSE to record production sequences, operating hours, door use, cleaning regimes and complaint locations.
- Produce measured HVAC as-built drawings in AutoCAD for the surveyed areas.
- Compare, for every zone, the rated capacity, the measured available capacity and the observed shortfall.

### 2.2 What you receive at the end of Stage 1

| Deliverable                  | Content                                                                     |
|------------------------------|-----------------------------------------------------------------------------|
| Survey and assessment report | Findings, condition, test results, complaint map and identified limitations |
| Measured asset register      | Every unit tagged, with make, model, capacity, area served and condition    |
| Performance test results     | Measured airflow, capacity, water and electrical readings per unit          |
| Monitoring data              | Raw logger files and processed temperature / humidity trend graphs by zone  |

| Deliverable       | Content                                                                      |
|-------------------|------------------------------------------------------------------------------|
| As-built drawings | HVAC zoning plan, equipment location plan, duct layouts with measured sizes. |

### 3. Stage 2 — Engineering Assessment and Detailed Design

Stage 2 turns the measured facts into an engineered solution, with the calculations to prove it and the documents needed to implement it.

#### 3.1 What we will do

- Hold a design criteria workshop with your Operations, QA/QC, Maintenance and HSE teams to agree the target temperature, humidity, fresh air and pressure requirement for every area. This becomes the approved design basis and is signed off before any equipment is selected.
- Prepare room-by-room heat and moisture load calculations using recognised engineering methods, showing sensible and latent loads separately. Calculations will include building fabric and solar gain, lighting, people, motors and process equipment, oven casing and radiant heat, proofer and steam moisture, hot product entering cooling and packaging areas, exhaust replacement air, door infiltration and envelope leakage.
- Prepare a complete area-by-area air balance and pressure cascade, quantifying exactly how much replacement air is missing and where.
- Carry out a root-cause analysis that links each complaint zone to measured evidence, distinguishing between genuinely insufficient capacity, equipment unable to deliver its rating, poor heat capture at source, exhaust and make-up air imbalance, excessive infiltration, poor air distribution, and incorrect control settings.
- Propose solutions in a fixed order of priority: restore existing systems first, reduce heat and moisture at source, correct make-up air and room pressure, reduce infiltration and improve air distribution, and only then add new cooling or ventilation capacity for the verified remaining shortfall.
- Propose either an optimized rectification option or a recommended permanent upgrade option — so that you can make an informed commercial decision.
- Select all equipment using certified manufacturer data corrected to the Qatar design ambient, with coils sized for both sensible and latent duty and fans sized at the calculated airflow and static pressure.
- Produce the detailed design: layouts, schematics, equipment schedules and selections, control philosophy, electrical loads and interfaces, and technical specifications.
- Identify the fire and life-safety provisions associated with the proposed modifications, including fire and smoke dampers, fire stopping, controls and fire-alarm interfaces, and include them in the design and BOQ.
- Prepare a detailed Bill of Quantities and the Stage 3 implementation proposal with the final project cost.

#### 3.2 What you receive at the end of Stage 2

| Deliverable                       | Content                                                          |
|-----------------------------------|------------------------------------------------------------------|
| Approved design criteria schedule | Agreed temperature, humidity, fresh air and ventilation per area |

| Deliverable                       | Content                                                                                      |
|-----------------------------------|----------------------------------------------------------------------------------------------|
| Load calculation report           | area-by-area loads, psychrometrics and supply air quantities                                 |
| Air balance                       | Area-by-area mass balance and proposed pressure cascade                                      |
| Root-cause report                 | Each complaint zone linked to measured evidence and proposed corrective action               |
| Options and recommendation report | Rectification option and upgrade option for each major intervention, with our recommendation |
| Detailed design drawings          | Layouts, schematics, equipment schedules, controls and electrical interfaces                 |
| Technical specifications          | Equipment, materials and workmanship specifications                                          |
| Bill of Quantities                | Priced-out quantities for the recommended works                                              |
| Stage 3 proposal                  | Detailed implementation proposal and final project cost                                      |

#### 4. Basis of Price — Surveyed Asset Schedule

Our price is based on surveying, tagging, condition-checking and performance-testing the following equipment population, derived from the equipment details issued with the tender for the bakery and oil building facilities.

| Asset group                                             | Quantity  | Remarks                        |
|---------------------------------------------------------|-----------|--------------------------------|
| Air handling units (bakery and oil building)            | 11        | AHU No. 3 excluded (cancelled) |
| Chillers (air-cooled and water-cooled)                  | 4         | Central plant                  |
| Cooling towers                                          | 2         | Central plant                  |
| Pumps (chilled water and condenser water)               | 6         | Central plant                  |
| Duct-type split, split and refrigeration units — bakery | 18        | Per A/C details schedule       |
| Exhaust and inlet fans — bakery                         | 32        | Per exhaust fan schedule       |
| <b>Total surveyed assets</b>                            | <b>73</b> |                                |

#### 5. What Is Included in Our Price

- All engineering, supervision, survey technicians and project management for Stages 1 and 2.
- All calibrated test instruments, with current calibration certificates supplied to you.
- Supply, installation, monitoring and removal of data loggers for the three-day monitoring period.
- Testing, adjusting and balancing for airflow and water measurement.
- All access equipment required for the survey — ladders, mobile towers, scissor lift, roof and ceiling access.
- All safety documentation: risk assessments, method statements, permits and PPE.
- Measured HVAC as-built drawings for the surveyed areas, in editable AutoCAD format.
- All calculations, reports, schedules and the BOQ, issued in editable native formats as well as PDF.
- Attendance at the design criteria workshop and at progress and review meetings.
- One round of revision to the Stage 1 and Stage 2 deliverables following your written comments.
- Handover of all information produced under this assignment as your project record.

#### 6. What Is Excluded from Our Price

The following are clearly excluded. Where an excluded item is later required, it will be quoted and proceed only with your written approval.

- Stage 3 works of any kind — supply, installation, repair, rectification, cleaning, replacement or modification of equipment. Stage 3 will be proposed separately after your approval of Stage 2.
- Authority submissions, approvals and fees (Qatar Civil Defence, Kahramaa, Ministry of Public Health) and the services of an authority-approved consultant. Confirmed as outside our scope by Tender Circular-1, Queries 4 and 7. Our design will comply with applicable Qatar regulations, codes and standards and will be suitable for future authority submission.
- Dust Hazard Analysis by a licensed specialist.

- Asbestos, hazardous material or environmental surveys of any kind.
- Architectural, structural or civil survey and drawings beyond the HVAC-relevant measured information required for this assignment.
- Intrusive opening, demolition, ceiling or cladding removal and subsequent reinstatement. Where required to trace concealed services, this will be identified, priced and carried out only with your written approval.
- Modification, testing, servicing or OEM attendance on process equipment such as ovens, proofers and production lines.
- Upstream electrical infrastructure studies beyond the boards and circuits directly serving the HVAC equipment. Transformer, main switchboard or major distribution assessment is excluded.
- CFD or advanced air-distribution modelling. Available as a priced option if required.
- Any repair, cleaning, replacement or adjustment of existing equipment during the survey, including filter changes and coil cleaning.
- Provision of temporary cooling, temporary ventilation or standby equipment.
- Extension of the monitoring period beyond three representative production days.
- Working outside the agreed survey hours, and working on public holidays or during unplanned production shutdowns.
- Repair or replacement cost arising from failure or leakage of existing equipment during or following testing. Existing equipment is tested in its as-found condition.
- Any duties, taxes or levies imposed after the date of this offer.

## 7. Assumptions and Clarifications

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1. The price is based on the surveyed asset schedule and the area boundary stated in Section 5 of this proposal.
2. The preliminary room criteria in Appendix B of the Scope of Work are used as the basis of this offer. The oven, proofing and product-cooling areas currently state 'ventilation required' with no numeric criterion. These will be fixed at the design criteria workshop. If criteria tighter than Appendix B are subsequently instructed, the resulting change in scope will be advised as a variation before we proceed.
3. Monitoring is priced for three representative production days as permitted by clause 6.5 of the Scope of Work. Extension to seven days, if instructed by you or required because results are inconclusive, will be charged at an agreed daily rate.
4. The survey requires access during at least one full representative production shift, plus off-shift access to reach equipment safely. Permitted survey hours are to be confirmed. Our price assumes access during normal working hours plus a reasonable allowance for off-shift inspection.
5. You will provide site access, a nominated site representative, electrical power for test instruments, and safe working conditions in accordance with your site HSE requirements.
6. The architectural drawings already issued are treated as reference information only. All dimensions and site conditions will be verified by MES, as confirmed in Tender Circular-1, Query 1.
7. Client review of each submission, and of each hold point, is assumed to be completed within ten working days. Delay beyond this period will extend the programme accordingly.
8. The performance obligation described in Section 13 of the Scope of Work applies to systems designed and installed by MES under Stage 3. It attaches only after Stage 3 is awarded, and only where the full set of approved recommendations is implemented, including recommendations relating to door discipline, envelope sealing, source capture and production practice. Where recommendations are implemented only in part, the performance guarantee cannot be maintained.
9. MES accepts no responsibility for the condition, performance or continued operation of existing equipment that is not modified under an awarded Stage 3 scope.
10. Items 8 and 9 of the tender Appendix C price schedule — testing, adjusting, balancing, commissioning, training, as-builts, O&M manuals and handover — relate to Stage 3 works and cannot be firmly priced before the Stage 2 design is approved. These will be quoted with the Stage 3 proposal.
11. All documents and data produced under this assignment become your property upon settlement of the related milestone payment.



## 8. Programme

| Activity                                                  | Duration        |
|-----------------------------------------------------------|-----------------|
| Mobilisation and approval of investigation plan           | 1 week          |
| Site survey, asset register, condition inspection         | 2 weeks         |
| Monitoring and performance testing (partly in parallel)   | 1 weeks         |
| Measured as-built drawing production                      | 2 weeks         |
| Stage 1 report compilation and submission                 | 1 weeks         |
| Client review and design criteria workshop (Hold Point 1) | 1 weeks         |
| Load calculations, air balance and root-cause analysis    | 1 weeks         |
| Options report, detailed design, specifications and BOQ   | 2 weeks         |
| <b>Total — Stage 1 and Stage 2 complete</b>               | <b>11 weeks</b> |

*The programme runs from the date of the Letter of Purchase Order and confirmed site access. Stage 3 procurement and installation duration will be advised with the Stage 3 proposal.*

## 9. Commercial Proposal

### 10.1 Price schedule

| Item                               | Description                                                                                                                             | Basis           | Amount (QAR)   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------|
| 1                                  | Stage 1 complete site survey, monitoring, asset register, performance testing and baseline as-built documentation                       | Lump sum        | 105,000        |
| 2                                  | Stage 2 design criteria workshop, load calculations, air balance, root-cause analysis, options, detailed design, specifications and BOQ | Lump sum        | 60,000         |
| <b>TOTAL — STAGE 1 AND STAGE 2</b> |                                                                                                                                         | <b>Lump sum</b> | <b>165,000</b> |

**Total lump sum price: QAR 165,000.00** (Qatari Riyals One Hundred and Sixty-Five Thousand only)

The price is fixed for the scope, asset schedule and assumptions stated in this proposal, and excludes any duties or taxes.

### 9.2 Milestone payment schedule

| No.          | Milestone                                                                                   | Trigger         | %           | Amount (QAR)   |
|--------------|---------------------------------------------------------------------------------------------|-----------------|-------------|----------------|
| M1           | Mobilisation — LPO received, investigation plan submitted and survey team mobilised to site | On mobilisation | 20%         | 33,000         |
| M2           | Site survey, monitoring and performance testing completed                                   | On completion   | 25%         | 41,250         |
| M3           | Stage 1 survey report, asset register and as-built drawings submitted                       | On submission   | 25%         | 41,250         |
| M4           | Stage 2 calculations, root-cause report, detailed design and BOQ submitted                  | On submission   | 20%         | 33,000         |
| M5           | Client approval of the Stage 2 design and issue of final documentation                      | On approval     | 10%         | 16,500         |
| <b>TOTAL</b> |                                                                                             |                 | <b>100%</b> | <b>165,000</b> |

*Invoices are payable within 30 days of submission. Milestone M5 will be invoiced 30 days after submission of the Stage 2 deliverables if client approval has not been issued by that date for reasons outside our control.*

## 10. Commercial Terms

|                        |                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <b>Offer validity</b>  | 60 days from the date of this proposal                                                                                     |
| <b>Payment terms</b>   | As per the milestone schedule in Section 10.3, payable within 30 days of invoice                                           |
| <b>Order</b>           | Letter of Purchase Order to be addressed to Metri Engineering Services W.L.L.                                              |
| <b>Delivery Time</b>   | 11 weeks from LPO and confirmed site access                                                                                |
| <b>Price basis</b>     | Fixed lump sum for the scope, asset schedule, assumptions and exclusions stated herein                                     |
| <b>Variations</b>      | Any change in scope, asset quantity, criteria or access conditions will be advised in writing and priced before proceeding |
| <b>Confidentiality</b> | All client information and production data obtained during the assignment will be treated as confidential                  |
| <b>Governing law</b>   | The laws of the State of Qatar                                                                                             |

We thank you for the opportunity to submit this proposal and remain available to discuss any aspect of it, or to attend a clarification meeting at your convenience.

**METRI ENGINEERING SERVICES W.L.L.**

